Second-Order Linear Homogeneous Equations

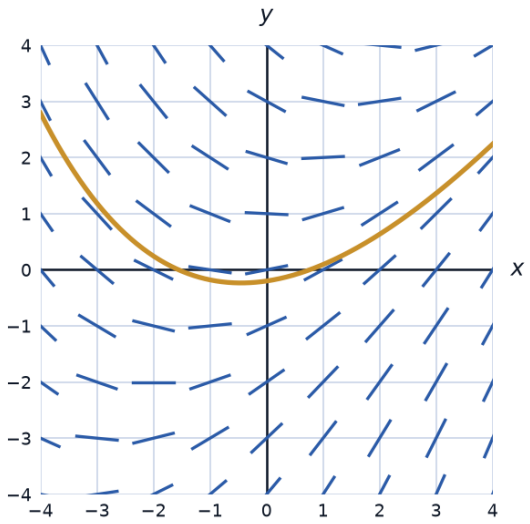
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: The auxiliary equation

Write the auxiliary (characteristic) equation for $y'' - 5y' + 6y = 0$.

2 [Foundational] • Target: The auxiliary equation

Write and simplify the auxiliary equation for $2y'' + 4y' - 6y = 0$.



3 [Foundational] • Target: The auxiliary equation

Explain why substituting $y = e^{rx}$ into $ay'' + by' + cy = 0$ produces the auxiliary equation $ar^2 + br + c = 0$.

4 [Foundational] • Target: Distinct real roots

Solve $y'' - 5y' + 6y = 0$.

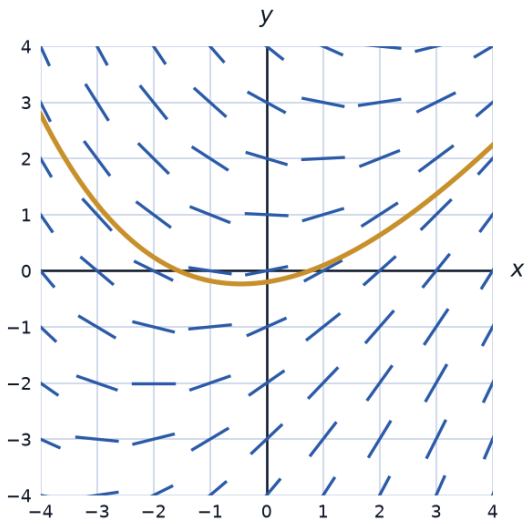
5 [Foundational] • Target: Distinct real roots

Solve $y'' + 2y' - 3y = 0, y(0) = 1, y'(0) = 1$.

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6 [Foundational] • Target: Distinct real roots

Solve $y'' - y' - 6y = 0$.



7 [Foundational] • Target: Distinct real roots

Solve $y'' - 9y = 0$ (note this factors as a difference of squares in r).

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Repeated roots

Solve $y'' - 6y' + 9y = 0$.

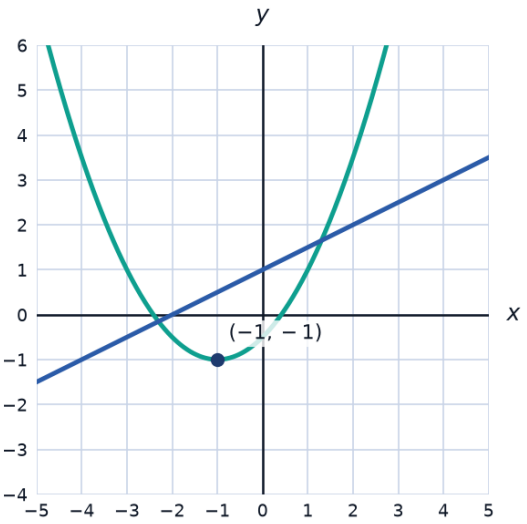
9 [Intermediate] • Target: Repeated roots

Solve $y'' + 4y' + 4y = 0$, $y(0) = 2$, $y'(0) = -1$.

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10 [Intermediate] • Target: Repeated roots

Explain why, for a repeated root r, $y_2 = xe^{rx}$ (not just a second copy of e^{rx}) is needed to build the general solution.



11 [Intermediate] • Target: Complex roots

Solve $y'' + 4y = 0$.

12 [Intermediate] • Target: Complex roots

Solve $y'' - 2y' + 5y = 0$.

13 [Intermediate] • Target: Complex roots

Solve $y'' + 9y = 0$, $y(0) = 2$, $y'(0) = 3$.

14 [Intermediate] • Target: Complex roots

Solve $y'' + 2y' + 2y = 0$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

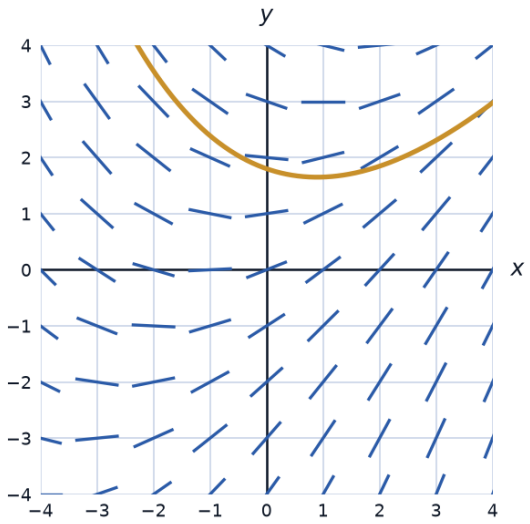
15 [Challenging] • Target: Classifying without fully solving • Cross-Lesson: 1.01 Separable Equations

Without solving, state whether $y'' + 6y' + 5y = 0$ has real, repeated, or complex roots, using the discriminant $b^2 - 4ac$ of the auxiliary equation.

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16 [Challenging] • Target: Classifying without fully solving • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Without solving, classify the roots of $y'' + 10y' + 25y = 0$.



17 [Challenging] • Target: Classifying without fully solving • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

Without solving, classify the roots of $y'' + 2y' + 10y = 0$.

18 [Challenging] • Target: Application: a damped spring • Cross-Lesson: 5.01 Variation of Parameters

A mass–spring–damper system satisfies $y'' + 4y' + 5y = 0$. Find the general solution and classify the damping (over/critically/under-damped).

19 [Challenging] • Target: Application: a damped spring • Cross-Lesson: 1.01 Separable Equations

Solve $y'' + 6y' + 9y = 0$, $y(0) = 1$, $y'(0) = 0$, and classify the damping.

20 [Challenging] • Target: Application: a damped spring • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Solve $y'' + 5y' + 6y = 0$, $y(0) = 1$, $y'(0) = 0$, and classify the damping.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Distinct Real Roots

Solve the auxiliary characteristic equation for $y'' - 5y' + 6y = 0$.

a. $r = 2$ and $r = 3$

b. $r = -2$ and $r = -3$

c. $r = 1$ and $r = 6$

d. $r = \pm\sqrt{6}$

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22 [Foundational] • Target: Distinct Real Roots

What is the general solution to $y'' - 5y' + 6y = 0$?

a. $y = c_1e^{2x} + c_2e^{3x}$

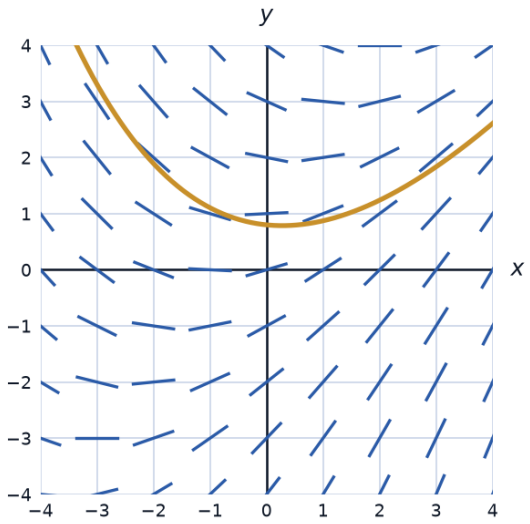
b. $y = c_1e^{-2x} + c_2e^{-3x}$

c. $y = (c_1 + c_2x)e^{2x}$

d. $y = c_1\cos(2x) + c_2\sin(3x)$

23 [Foundational] • Target: Distinct Real Roots

Solve the IVP $y'' - y = 0$ with $y(0) = 4$ and $y'(0) = 2$.



a. $y = 3e^x + e^{-x}$

b. $y = 2e^x + 2e^{-x}$

c. $y = 4e^x$

d. $y = e^x + 3e^{-x}$

24 [Foundational] • Target: Repeated Real Roots

What is the general solution to $y'' - 6y' + 9y = 0$?

a. $y = (c_1 + c_2x)e^{3x}$

b. $y = c_1e^{3x} + c_2e^{-3x}$

c. $y = c_1e^{3x} + c_2x^2e^{3x}$

d. $y = c_1\cos(3x) + c_2\sin(3x)$

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25 [Foundational] • Target: Repeated Real Roots

Why is the factor x introduced into the second solution xe^{rx} for repeated roots?

a. To ensure linear independence from the first solution e^{rx}

b. Because the derivative of e^{rx} contains x

c. To make the solution satisfy $y(0) = 0$

d. By Euler's identity

26 [Foundational] • Target: Repeated Real Roots

Solve $y'' + 4y' + 4y = 0$ with $y(0) = 2$ and $y'(0) = -1$.

a. $y = (2 + 3x)e^{-2x}$

b. $y = (2 - x)e^{-2x}$

c. $y = 2e^{-2x}$

d. $y = (2 + x)e^{-2x}$

27 [Foundational] • Target: Complex Conjugate Roots

If the characteristic equation has complex roots $r = \alpha \pm i\beta$, what is the real general solution?

a. $y = e^{\alpha x}(c_1\cos(\beta x) + c_2\sin(\beta x))$

b. $y = c_1e^{\alpha x} + c_2e^{\beta x}$

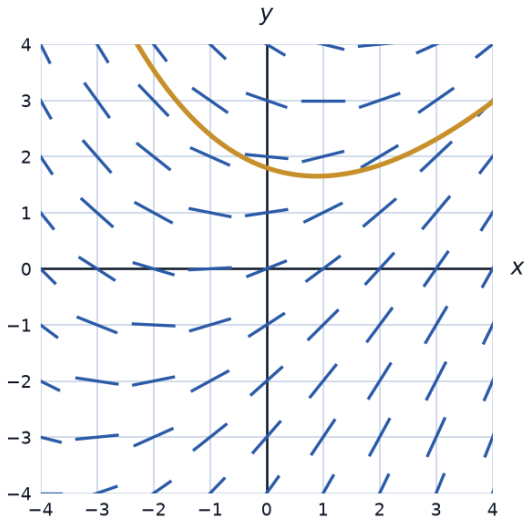
c. $y = e^{\beta x}(c_1\cos(\alpha x) + c_2\sin(\alpha x))$

d. $y = c_1\cos((\alpha + \beta)x)$

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28 [Foundational] • Target: Complex Conjugate Roots

Find the general solution to $y'' + 16y = 0$.



a. $y = c_1\cos(4x) + c_2\sin(4x)$

b. $y = c_1e^{4x} + c_2e^{-4x}$

c. $y = (c_1 + c_2x)e^{4x}$

d. $y = c_1\cos(16x) + c_2\sin(16x)$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Complex Conjugate Roots

Solve the characteristic equation for $y'' - 2y' + 5y = 0$.

a. $r = 1 \pm 2i$

b. $r = 2 \pm 4i$

c. $r = -1 \pm 2i$

d. $r = 1 \pm 4i$

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30 [Intermediate] • Target: Harmonic Motion and Damping Regimes

A mechanical oscillator modeled by $my'' + cy' + ky = 0$ is underdamped when:

a. $c^2 - 4mk < 0$ (oscillatory motion with exponentially decaying amplitude)

b. $c^2 - 4mk = 0$ (critically damped)

c. $c^2 - 4mk > 0$ (overdamped, no oscillations)

d. $c = 0$ (undamped simple harmonic motion)

31 [Intermediate] • Target: Harmonic Motion and Damping Regimes

Which damping condition returns a displaced mass to equilibrium in the shortest time without oscillating?

a. Critical damping ($c^2 = 4mk$)

b. Overdamping ($c^2 > 4mk$)

c. Underdamping

d. Zero damping

32 [Intermediate] • Target: Harmonic Motion and Damping Regimes

What is the natural frequency ω_0 of the undamped oscillator $y'' + 25y = 0$?

a. $\omega_0 = 5$ rad/s

b. $\omega_0 = 25$ rad/s

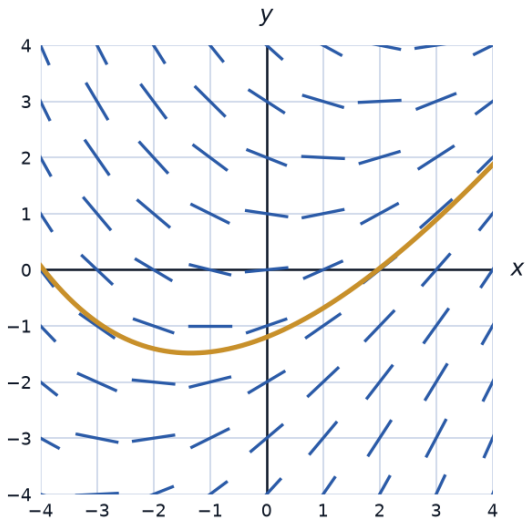
c. $\omega_0 = 10$ rad/s

d. $\omega_0 = 2.5$ rad/s

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33 **[Intermediate]** • **Target: The auxiliary equation**

Consider the second-order linear homogeneous differential equation: $y''(x) - ky'(x) + 4y(x) = 0$. If the characteristic equation has a repeated real root, and one of the solutions to the differential equation is $y_1(x) = xe^{2x}$, determine the value of the parameter k.



a. 2

b. 4

c. 3

d. 1

34 **[Intermediate]** • **Target: Distinct real roots**

Consider the second-order linear homogeneous differential equation $y'' - 2y' + y = 0$. \\ If one of the fundamental solutions is $y_1(x) = e^x$, and the Wronskian of two linearly independent solutions $y_1(x)$ and $y_2(x)$ is given by $W(y_1, y_2)(x) = 5e^{2x}$, find the unknown parameter k in $y_2(x) = kxe^x$.

a. $k = 5$

b. $k = -5$

c. $k = 1$

d. $k = -1$

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35 **[Intermediate]** • **Target: Repeated roots**

Consider the second-order linear homogeneous differential equation $y'' - 2y' + ky = 0$. We are given that $y_1(x) = e^{3x}$ is a solution to this equation. \\ (a) Find the value of the parameter k. \\ (b) Find the general solution to the differential equation.

a. $k = -3, y(x) = c_1e^{3x} + c_2e^{-x}$

b. $k = 3, y(x) = c_1e^{3x} + c_2xe^{3x}$

c. $k = -3, y(x) = c_1e^{3x} + c_2e^{x}$

d. $k = 9, y(x) = c_1e^{3x} + c_2e^{-3x}$

36 **[Intermediate]** • **Target: Complex roots**

Consider the second-order linear homogeneous differential equation $y'' - 3y' + 2y = 0$. Find the general solution to this equation.

a. $y(x) = c_1e^{x} + c_2e^{2x}$

b. $y(x) = c_1e^{-x} + c_2e^{-2x}$

c. $y(x) = c_1\cos(x) + c_2\sin(x)$

d. $y(x) = c_1xe^{x} + c_2e^{x}$

37 **[Intermediate]** • **Target: Classifying without fully solving**

Consider the second-order linear homogeneous differential equation $y'' + P(x)y' + Q(x)y = 0$. For the equation $x^2y'' + xy' + (x^2 - u^2)y = 0$ (Bessel's equation of order u), the point $x = 0$ is a regular singular point. If we assume a series solution of the form $y(x) = \sum_{n=0}^{\infty} a_nx^{n+r}$, where a_0eq0, the indicial equation is given by $r^2 - u^2 = 0$. For this problem, let $u = 1$, so the indicial equation is $r^2 - 1 = 0$. This yields two distinct roots, $r_1 = 1$ and $r_2 = -1$. For the larger root $r_1 = 1$, the power series solution takes the form $y_1(x) = x + \sum_{n=1}^{\infty} a_nx^{n+1}$. Substituting this into Bessel's equation of order 1, and after considerable algebraic manipulation, the recurrence relation for the coefficients a_n is found to be $a_n = -\frac{a_{n-2}}{n(n+2)}$ for $n \geq 2$. Given $a_0 = 1$, find the value of a_3 for the solution corresponding to the larger root $r_1 = 1$.

a. 0

b. 1/24

c. 1/48

d. 1/12

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Application: a damped spring • Cross-Lesson: 5.01 Variation of Parameters

Consider the second-order linear homogeneous differential equation $y'' - 5y' + 6y = 0$. If the auxiliary equation has roots r_1 and r_2, find the value of $r_1 + r_2$.

a. 5

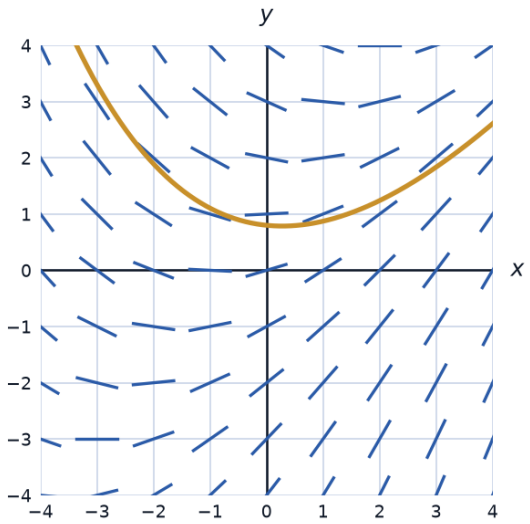
b. -5

c. 6

d. -6

39 [Challenging] • Target: Verifying general solutions and the superposition principle • Cross-Lesson: 1.01 Separable Equations

Consider the second-order linear homogeneous differential equation $y''(x) + 5y'(x) + 6y(x) = 0$. If the auxiliary equation has roots r_1 and r_2, what is the value of $r_1 + r_2$?



a. 5

b. -5

c. 6

d. -6

40 [Challenging] • Target: Boundary value problems • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Consider the second-order linear homogeneous differential equation $y'' - 5y' + 6y = 0$. If the characteristic equation has roots r_1 and r_2 with $r_1 < r_2$, what is the value of r_1?

a. 1

b. 2

c. 3

d. 6

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41 [Challenging] • Target: Distinct Real Roots • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

For which values of k does every nonzero solution of $y'' + 4y' + ky = 0$ genuinely oscillate (the auxiliary equation has non-real roots)?

a. $k < 4$

b. $k = 4$

c. $k > 0$

d. $k > 4$

42 [Challenging] • Target: Repeated Real Roots • Cross-Lesson: 5.01 Variation of Parameters

Consider the second-order linear homogeneous differential equation: $x^2y'' + xy' + (x^2 - u^2)y = 0$, where u is a constant. This is Bessel's equation of order u. Assume $u = 1/2$ and that $x > 0$. x is a regular singular point. Determine the indicial equation for this differential equation and find its roots.

a. Indicial equation: $r^2 - \frac{1}{4} = 0$; roots: $r = \pm\frac{1}{2}$

b. Indicial equation: $r^2 - r + \frac{1}{4} = 0$; roots: $r = \frac{1}{2}, \frac{1}{2}$

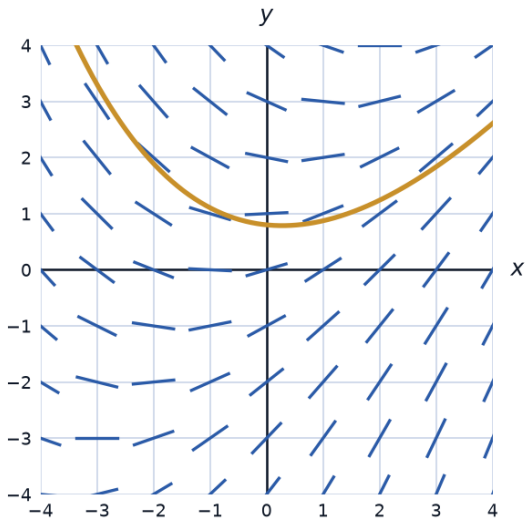
c. Indicial equation: $r^2 - \frac{1}{4} = 0$; roots: $r = \pm\frac{1}{4}$

d. Indicial equation: $r^2 + \frac{1}{4} = 0$; roots: $r = \pm\frac{i}{2}$

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43 [Challenging] • Target: Complex Conjugate Roots • Cross-Lesson: 1.01 Separable Equations

Consider the second-order linear homogeneous differential equation: $y'' - xy' + y = 0$. The point $x = 0$ is an ordinary point. Find the first two non-zero terms of the power series solution around $x = 0$, assuming $y(0) = 1$ and $y'(0) = 0$.



a. $1 + \frac{1}{2}x^2 + \ldots$

b. $1 + \frac{1}{3}x^3 + \ldots$

c. $1 - \frac{1}{2}x^2 + \ldots$

d. $1 - \frac{1}{3}x^3 + \ldots$

44 [Challenging] • Target: Harmonic Motion and Damping Regimes • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Consider the nonhomogeneous second-order linear differential equation $y'' - y = \frac{1}{1+e^x}$. If $y_1(x) = e^x$ and $y_2(x) = e^{-x}$ are two linearly independent solutions to the associated homogeneous equation, find the Wronskian of y_1 and y_2.

a. 2

b. -2

c. 0

d. $e^x - e^{-x}$

45 [Challenging] • Target: The auxiliary equation • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

A damped spring satisfies $y'' + 2y' + 10y = 0$, $y(0) = 0$, $y'(0) = 6$. Find $y(\pi/6)$.

a. $2e^{\pi/6}$

b. $2e^{-\pi/6}$

c. $-2e^{-\pi/6}$

d. $e^{-\pi/6}$

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